

1 Deploy to MCU

Trained CNN
Conv + BatchNorm (fp32)

fold BN +
int8 quantize

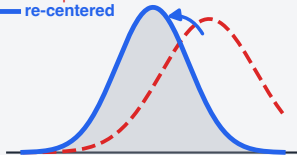
Integer-only model
BN fused into Conv

adaptation no longer works

0 points recovered

2 Recalibrate per channel

■ clean target
- - corrupted
— re-centered



per-channel pre-activation
match each channel to its clean (β , $|\gamma|$)
forward-only · no gradients

3 Run on ESP32-S3

accuracy recovered (pts)

■ deployed +0.0

+20.9

FORGE

energy +8.3 mJ (6.8%)

latency +21.9 ms (7.6%)

measured, Nordic PPK2